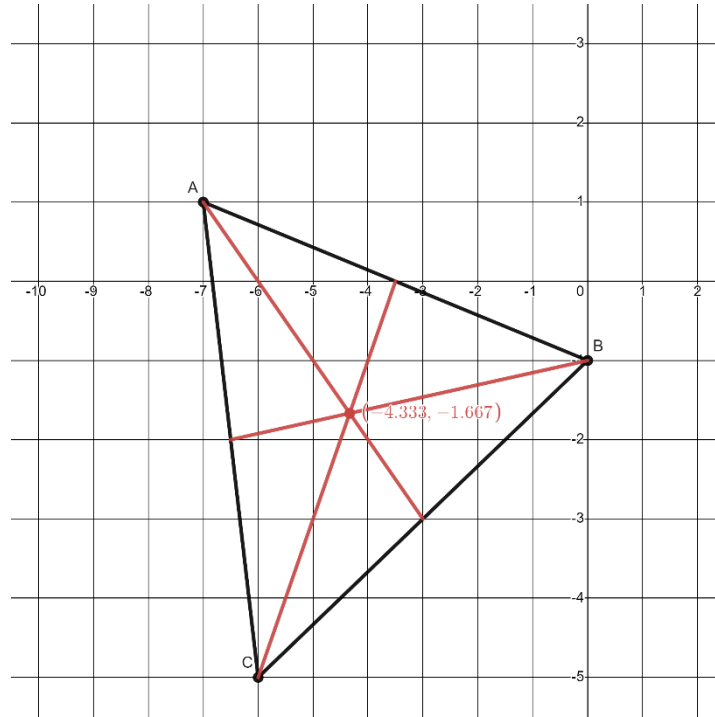


Geometry
Unit 13
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

Use the following information for questions 1-4. $\triangle ABC$ has vertices at $A(-7,1)$, $B(0,-1)$, and $C(-6,-5)$.



1. What is the average of the x -coordinates? Round to the nearest thousandth if necessary.

$$\frac{-7+0+(-6)}{3} = -4.333$$

2. What is the average of the y -coordinates? Round to the nearest thousandth if necessary.

$$\frac{1+(-1)+(-5)}{3} = -1.667$$

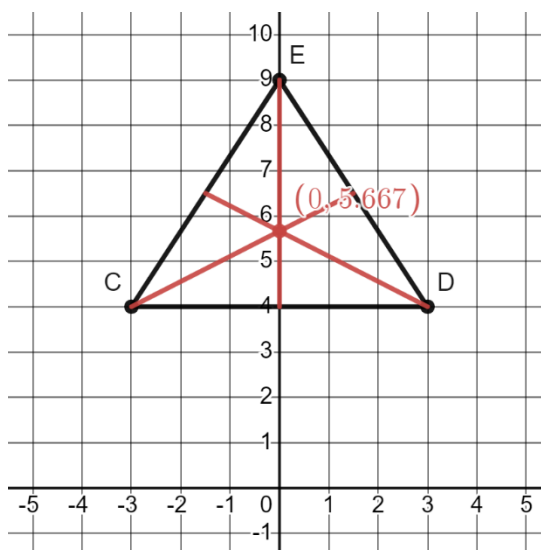
3. What are the coordinates of the centroid?

$$(-4.333, -1.667)$$

4. Verify the coordinates of the centroid by drawing the median of each side.

See graph.

$\triangle CDE$ has vertices at $C(-3,4)$, $D(3,4)$, and $E(0,9)$.



5. What is the average of the x -coordinates? Round to the nearest thousandth if necessary.

$$\frac{-3+3+0}{3} = 0$$

6. What is the average of the y -coordinates? Round to the nearest thousandth if necessary.

$$\frac{4+4+9}{3} = 5.667$$

7. What are the coordinates of the centroid of $\triangle CDE$?

$(0, 5.667)$

8. Verify the coordinates of the centroid by drawing the median of each side.

See graph.

$\triangle NPQ$ has vertices at $N(8,3)$, $P(9,-3)$, and a centroid at $(\frac{10}{3}, \frac{4}{3})$.

9. Graph $\triangle NPQ$ on the coordinate plane.

See graph.

10. What are the coordinates of vertex Q ?

$$\begin{aligned} \frac{8+9+x}{3} &= \frac{10}{3} \\ 17+x &= 10 \\ x &= -7 \\ \frac{3+(-3)+y}{3} &= \frac{4}{3} \\ y &= 4 \\ (-7, 4) \end{aligned}$$

